

Practice — Electromagnets

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

A current-carrying coil that behaves as a magnet is called an —

- (a) electrolyte (b) electromagnet (c) electrode (d) electron

Q2 • MCQ**[1]**

To make an electromagnet stronger you should —

- (a) use fewer turns (b) use a single cell (c) increase the current and number of turns
(d) remove the iron core

Q3 • MCQ**[1]**

An electromagnet has —

- (a) only a North pole (b) only a South pole (c) two poles, North and South (d) no poles

Q4 • TRUE / FALSE**[1]**

An electromagnet with a battery of 2 cells attracts more clips than the same one with a single cell.

- (a) True (b) False

Q5 • ASSERTION–REASON**[1]**

A: Most electromagnets have an iron core.

R: An iron core makes the electromagnet much stronger.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

The poles of an electromagnet can be reversed by reversing the _____ of the current.

Q7 · FILL IN THE BLANK

[1]

Strong electromagnets hung from cranes to lift scrap metal are called _____ electromagnets.

Short answer (2 marks each)

Q8

[2]

What is an electromagnet? How can it be made stronger?

Q9

[2]

Give two advantages of an electromagnet over a permanent bar magnet.

Q10

[2]

What happens to the iron paper clips held by an electromagnet when the current is switched off?
Why?

Long / application (3 marks each)

Q11

[3]

Sumana's lifting-electromagnet model picks up clips when the circuit is closed and drops them when opened. Explain how it works and why it stops lifting if the cell becomes weak.

Q12

[3]

Neha keeps the coil the same but slides the iron nail out. Will the coil still deflect the compass? Will the deflection be more or less than before? Explain.

[Check yourself](#) → Answer key